

Fake News Detection



By

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60180

BSDS 5-1

Subject

Data Mining

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Spring – 2026

1. Introduction

As online platforms grow, the question of misinformation has become all too real. Fake News can be misused, create confusion and play a role on mass-scale influence to public opinion. The goal of this project is to create a classifier that can be trained using machine learning(ML) and with the help of it evaluate whether any news article is real or fake.

2. Dataset Overview

The dataset consists of two separate CSV files that were merged into a unified dataset for analysis:

- **Fake.csv** — containing fake news articles
- **True.csv** — containing real news articles

Each file includes the title, full text/content, subject/category, and publication date. After loading, both datasets were merged, labels were assigned (Fake = 0, Real = 1), and the dataset was shuffled to ensure randomness and avoid bias during model training.

3. Data Preparation

To ensure high-quality results, the dataset was passed through several preprocessing steps:

a. Cleaning

- Removed duplicate records
- Filled missing values in text and title fields
- Standardized column formats

b. Text Processing

Since the data is textual, it required the following cleaning steps:

- Converted all text to lowercase
- Removed URLs, special characters, and unnecessary symbols
- Removed common stop words (e.g., "the", "and")
- Applied lemmatization to reduce words to their base form

c. Feature Engineering

- **Combined title + text** into a single column for richer context
- **TF-IDF Vectorization** — converted text into numerical format using Term Frequency–Inverse Document Frequency, which highlights words that best distinguish fake from real news

4. Exploratory Data Analysis (EDA)

Several visualizations were created to understand the dataset before modelling:

a. Label Distribution

- Bar chart and pie chart confirmed the dataset is balanced
- Ensures unbiased model training

b. Subject Distribution

- Identified the most common topics in the dataset
- Showed how fake and real news differ by subject category

c. Word Clouds

- Fake news showed more sensational, emotional, and politically charged terms
- Real news showed more formal, structured, and factual language

d. Text Length Analysis

- Compared article length between fake and real news
- Real news articles were generally more concise and structured

5. Model Development

The dataset was split into three parts to ensure reliable and generalizable evaluation:

- **70% Training** — 31,428 articles used to teach the models
- **15% Testing** — 6,703 articles used to measure performance on unseen data
- **15% Validation** — 6,767 articles used as an independent second check

Six machine learning models were trained and evaluated:

- I. **Naive Bayes:** Fast and efficient for text classification; performs well with TF-IDF features.
- II. **Random Forest:** An ensemble of decision trees that captures complex patterns in data.
- III. **Logistic Regression:** A simple yet powerful linear model well-suited to high-dimensional text data.
- IV. **Support Vector Machine (SVM):** Highly effective for text classification with strong generalization.
- V. **K-Nearest Neighbors (KNN):** Classifies based on similarity; required dimensionality reduction for efficiency.
- VI. **DBSCAN (Clustering):** An unsupervised method used to explore natural groupings in the data.

6. Model Evaluation

Each model was evaluated using five standard metrics: Accuracy, Precision, Recall, F1-Score, and AUC-ROC. The table below summarizes the key results:

Model	Accuracy	F1-Score	AUC-ROC	Remarks
Naive Bayes	94.85%	94.55%	98.76%	Fast, efficient
Random Forest	98.81%	98.78%	99.90%	Strong & robust
Logistic Regression	99.25%	99.21%	99.96%	Most balanced
SVM ★ Best	99.36%	99.32%	99.96%	Highest overall
KNN	94.60%	94.41%	98.38%	Good but slower

Key Findings

- All classification models achieved accuracy above 94%
- SVM achieved the highest overall performance: 99.36% accuracy, 99.32% F1-Score, and 0.9996 AUC-ROC
- Logistic Regression was the most balanced performer with consistent results across all metrics
- Naive Bayes was the fastest model, making it suitable for real-time applications
- Confusion matrices showed very few misclassifications across all models
- ROC curves for all models curve sharply toward the top-left, far above the random baseline

DBSCAN Clustering

- All articles formed a single dense cluster with no noise points
- Confirms that supervised classification is necessary — the two classes cannot be separated by simple grouping alone

7. Model Comparison

- **Best Model:** SVM (highest F1-Score and AUC-ROC)
- **Most Balanced:** Logistic Regression
- **Fastest:** Naive Bayes
- **Most Computationally Heavy:** Random Forest and KNN
- **Exploratory Only:** DBSCAN — useful for data understanding, not prediction

8. Visualizations Summary

The project produced the following visual outputs to support analysis and communicate results clearly:

- Label distribution charts (bar and pie)
- Subject/category analysis by label
- Word clouds for fake vs. real news
- Text length comparison (box plots)
- Confusion matrices for each model (test and validation sets)
- Model comparison bar chart across all metrics
- Performance metrics heatmap
- ROC curve comparison for all 5 classifiers
- KNN optimization graph (K-value vs. validation accuracy)
- DBSCAN clustering visualization (2D projection)

9. Conclusion

This project proves that machine learning can successfully identify fake news from text. The combination of extensive data preprocessing, TF-IDF feature extraction along with six machine learning models led to high model accuracy together with strong performance on most metrics.

The Support Vector Machine (SVM) gave the best and most consistent performance of any model, achieving 99.36% accuracy and a near-perfect AUC-ROC of 0.9996 across all iterations. Logistic Regression Trailing Not Far Behind as Second Most Stable Performer Yet, the solution is a solid, production ready solution that can be deployed in real world applications to help mitigate misinformation.